Yiqun Zhou

z-yq@live.cn | github.com/Roswellii | roswellii.github.io | Hangzhou, China

Research Interests

My research goal is to make robots solve real problems safely and reliably.

Embodied AI and vision-and-language navigation, world models and predictive representations, 3D perception and SLAM. Years of deploying autonomous systems on real machines convinced me that the hard part of embodiment is not perceptual accuracy but dependability — whether a capability demonstrated once still holds after conditions change.

Education

2023 – present **Zhejiang University**, College of Computer Science and Technology
Ph.D. student (direct-entry), Computer Science and Technology · Intelligent Vehicle Research Center
Advised by Prof. Zhijie Pan

2019 – 2023 **Chongqing University**, College of Computer Science
B.Eng., Computer Science and Technology (Excellence Program) · CET-6: 580

Publications and Patents

Under review

1. **Yiqun Zhou**, Xiao Liu, Yuechen Wu, Junjie An, Jiaxiang Zhou, Yizheng Huang. *RouteGuide: Fast VLN Adaptation from a Successful Video.* ACL ARR 2026.
   VLN-CE val-unseen: SR 32.0% → 40.0%, SPL 0.213 → 0.275, NE 8.21 → 7.69.
2. **Yiqun Zhou**, Jiaxiang Zhou, Yizheng Huang, Xiao Liu, Yuechen Wu, Feixiang Jing. *Imagine, Interpret, Act: Giving Temporal Latents a Sense of Progress.* AAAI 2027.
   Closed-loop success improves by 4.7–14.7 points over a matched LeWM control across four visual-control tasks.
3. **Yiqun Zhou**, Jiaxiang Zhou, Yizheng Huang, Xiao Liu, Yuechen Wu. *TopoMoE: Topology-Aware Expert Routing for Mixture-of-Experts Inference over Mobile Edge Networks.* ACM MobiCom 2027.

Published

4. **Yiqun Zhou**, Tengfei Hu, Xiao Zhou, Pan Lv, Zhijie Pan, Hong Li, Guoqing Yang. *CurbMapping: Reducing Blind Spots for Low-cost Autonomous Sweepers.* CAC 2025.
5. **Yiqun Zhou**, Huimin Zhao, Giandomenico Caruso. *Dedicated Lanes for Intelligent and Connected Vehicles in Mixed Traffic.* CIVS 2025.
6. Shengpu Zhang, **Yiqun Zhou**, Pan Lv, Guoqing Yang, Hong Li, Zhijie Pan. *Loosely Coupled LiDAR-Inertial-Encoder SLAM for Large-Scale Environments with Limited FoV.* CAC 2025.
7. Chao Liu, Runfeng Cai, **Yiqun Zhou**, Xin Chen, Haibo Hu, Meng Yan. *Understanding the Implementation Issues When Using Deep Learning Frameworks.* Information and Software Technology, 166, 2024. CCF-B
8. Guanyu Chen, **Yiqun Zhou**, Guoqing Yang, Hong Li, Pan Lv*. *SmartKit: User-Friendly Robot with Multiple Operating Systems.* IROS 2024. CCF-C
9. Lan Gao, **Yiqun Zhou**, Xin Chen, Runfeng Cai, Guo Chen, Chaojie Li. *Privacy-Preserving Dynamic Average Consensus via Random Number Perturbation.* IEEE Trans. on Circuits and Systems, 2022.

Patents Four invention patents (1 granted, 2 under substantive examination, 1 filed): HD map construction for unmanned vehicles in large outdoor scenes; LiDAR blind-spot obstacle avoidance for low-speed outdoor sweepers; behaviour planning on structured pavement; single-lane vehicle routing under dynamic demand.

Engineering Projects

Deploying VLA policies and vision-language navigation on real robots
Anker Innovations · Internship · 2025 – present

I built the full data-collection and deployment rig, including a PICO headset teleoperation setup for demonstrations and the localization module. With $\pi_{0.5}$ I completed cloth-folding and related manipulation tasks on Piper and Franka arms, and implemented natural-language visual navigation on a Unitree Go2, covering data collection, model deployment and closed-loop validation on the physical robots.

Techinao RoboNeo — outdoor autonomous cleaning vehicles
Wuxi Techinao Intelligent Technology (founded by our team) · techinao.com · Team lead · 2022 – present

I built the vehicle software stack from zero, spanning embedded control, sensor selection and integration for cameras and LiDAR (Mid-360, RTK), and the mapping, localization, navigation and planning modules. I deployed a MoGe-2-based monocular depth estimation system on the vehicle, and led the team through several vehicle generations into production. The RoboNeo Max / Pro / C99 line now runs daily in scenic areas, shopping malls and factory warehouses.

Autonomous loader and sweeper at Lanshan Port, Shandong

Shandong Port Group project · 2024

Mapping and localization at a working bulk-cargo port, deploying an unmanned tracked loader to clean along a three-kilometre spoil conveyor. Structural features are sparse, dust and vibration are constant, operation cannot be interrupted, and the three-kilometre run makes trajectory drift the dominant error source, so long-run stability matters more than instantaneous accuracy. I built the map offline with FAST-LIO, added a keyframe strategy to suppress drift, and used NDT registration for global relocalization so that the system recovers its pose quickly at start-up and after any interruption; deployed and tuned on site.

Tianmen Mountain autonomous vehicle challenge — vehicle control

2025

Responsible for vehicle control, adapting and tuning openpilot's lateral and longitudinal control on a real car. The course is the Tianmen Mountain summit road: 99 hairpin turns and 1,100 m of vertical drop, with intermittent satellite reception and limited visibility.

Research Projects

Key technologies for vehicle-control operating systems in intelligent connected vehicles

NSFC Joint Fund for Regional Innovation and Development · Beihang University · Zhejiang University as collaborating institution · mid-term review March 2025

I contributed to the research on SmartKit, a mixed-criticality system for automotive scenarios: virtualization provides parallel execution and safety isolation across multiple operating systems, a real-time OS monitors the Linux guest, and on failure the system switches rapidly to a preloaded backup VM; ROS recovery was validated on an unmanned vehicle platform, published at IROS 2024. I prepared the mid-term review materials and presented as the Zhejiang University representative.

Automated test and verification toolchain for automotive ECU software

National Key R&D Program of China, "New Energy Vehicles" key special project · Zhejiang University as participating institution

I led the ZJU student team through system development, designing the ECU inspection platform and the ASAM protocol layer and completing integration with HIL rigs. I wrote the mid-term and final defense materials and led the acceptance demonstration.

Research Capabilities

Perception — I carry mapping, relocalization and calibration through the full pipeline in large real environments and keep them stable over long runs where structural features are sparse and interference is heavy — LiDAR-inertial odometry and NDT registration, point cloud semantic segmentation, multi-view geometry and camera calibration, RTK positioning and multi-sensor fusion.

Policies — I run the full VLA pipeline — data collection, training, inference and deployment on a real robot — and am familiar with mainstream VLA architectures and the action-chunking mechanism, as well as the design and evaluation of world models. I also know the mainstream computer-vision algorithms and am comfortable with training and deployment pipelines on the common compute platforms; related work includes closed-loop VLN evaluation, reinforcement learning, and VLM fine-tuning and distillation.

Systems — I take systems end to end, from sensor and algorithm integration to whole-machine deployment. I am familiar with ROS 1 / ROS 2 and its debugging toolchain, and with a range of robot platforms — Unitree Go2 and G1, Franka and Piper arms. I have delivered in the field on unmanned vehicles, manipulators and legged robots, with experience in edge-side inference optimization and real-time tuning.

Awards

2024 — Outstanding Student, Zhejiang University · Head of Academic Affairs, Ph.D. Student Union

2022 — **National Scholarship**, Ministry of Education · Huawei "Rising Star" Scholarship · Third Prize, Huawei ICT Competition (National) · Third Prize, National Embedded Chip Design Competition · Outstanding Completion, National Undergraduate Innovation Program

2020 – 2022 — Top-10 Youth League Member and Outstanding Student, Chongqing University